LYDIA PEDERSEN

AI PIPELINE ARCHITECT & DOCUMENTATION ENGINEER

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PROFESSIONAL SUMMARY

Documentation engineer and AI pipeline architect with 20 years of experience across physics research, hardware validation, technical writing, and AI-powered content systems. I am driven by the challenge of translating ambiguous human intention into highly controlled, quasi-deterministic programmatic workflows. I write in-depth technical content for engineering audiences, interview engineers to surface the decisions behind complex systems, and build agentic pipelines that replace manual judgments with reliable, automated systems. My pipelines rely on A/B testing, structured result auditing, cross-file consistency checks, and strict reproducibility, applying the same engineering discipline to AI systems that hardware teams apply to certification.

SELECTED ENGINEERING PROJECTS

Transformer Architecture & AI Foundations Ebook Series — Independent · 2026

- Authored five interconnected series (53 chapters, 5 ebooks) that build a decoder-only Transformer from first principles: BPE tokenization, embeddings, positional encoding (sinusoidal and RoPE), the forward and backward pass (Q/K/V attention, residual stream, layer normalization, MLP), and Mixture of Experts routing, with the tensor algebra derived by hand.

Signal Integrity Engineering Series & Ebook — Independent · 2026

- Authored a 21-chapter series and ebook spanning transmission lines, S-parameters, jitter, equalization, PAM4, SerDes, CDR, and coherent optics. A full revision corrected 60+ factual errors, derived every governing equation step by step, and enforced writing standards with a custom Python linter.

SERP API Benchmark for AI Overviews (Open Source) — Independent · 2026

- Built and open-sourced a Python and Playwright benchmark of how reliably four SERP APIs extract Google AI Overviews for RAG pipelines. Rebuilt the methodology around detection yield after Google's non-determinism invalidated one-to-one scoring, and audited 196 raw responses for silent parser failures.

AI Tooling Ecosystem & Reference Architecture Guides — Independent · 2026

- Authored a 6-part series mapping the AI tooling landscape (11 tool types and the decision rules for assembling them) and 5 reference architectures with setup guides and annotated configurations, spanning LangChain, LangGraph, LlamaIndex, Weaviate, Pinecone, Qdrant, vLLM, SGLang, Ollama, LiteLLM, Langfuse, and Braintrust.

PROFESSIONAL EXPERIENCE

Senior Technical Writer — Jun 2023 – Jul 2026

Gravitee · Remote

TECHNICAL WRITING

- Wrote and maintained developer documentation for an API management platform, including Kafka and Server-Sent Events integration guides, the expression language reference, Docker deployment, and gateway resource sizing. Interviewed engineers regularly to capture design decisions and tested configuration examples hands-on before publication.

AGENTIC PIPELINE DEVELOPMENT

- **AI Doc Writer Agent:** Multi-stage pipeline that converted PR diffs and Jira tickets into publication-ready documentation, with fidelity scoring, Best-of-N evaluation with adaptive scaling, a guard cascade against hallucination and editorial fabrication, and multi-stage recovery. Measured output quality with deduplication metrics and cross-file consistency checks.
- **Analytics Deflection Engine:** Refactored a monolith into a modular, product-agnostic pipeline that turned raw AI assistant analytics into ranked, privacy-safe documentation improvement reports, using a parameter sweep for self-correction, multi-signal priority scoring (impact, pain, breadth), a k-anonymity privacy model, and automated Jira task generation.
- **Link Governance Engine:** Multi-category link classifier with safety-gated autofix, page-version index validation, and external link health checks that automated broken link remediation across versioned documentation sites with auditable decisions.
- **Semantic Redirect Engine:** Extract, Embed, Match, Emit pipeline that used embedding similarity, a penalty stack, and a multi-tier fallback cascade to generate cross-version page mappings that preserved user navigation across releases.

AI DEVELOPMENT METHODOLOGY

- Built all pipelines through AI-assisted development with agentic and prompt engineering patterns, model selection tuned to token budgets and rate limits, and checkpoints and structured handoffs that preserved context across sessions.

DOCUMENTATION ENGINEERING & STRATEGY

- Automated headless browser screenshot capture, eliminating manual image maintenance. Elevated documentation to a first-class engineering deliverable by coauthoring team processes, templates, and roadmaps, and redesigned site information architecture around user journeys.

Technical Writer Oct 2021 – Mar 2023

Entando · Remote

Owned developer documentation for a Kubernetes-based application composition platform. Verified all content in cloud and Multipass VM environments with hands-on testing of MFE/MS generation, product UIs, and CLI tooling.

- Maintained a VuePress static documentation site with Markdown/Vue styling. Created and owned the company glossary and style guide, standardizing terminology across all customer-facing materials.
- Authored a white paper used to secure new investors. Edited marketing, developer advocacy, and partner communications for content and language. Coached non-writers in documentation best practices.

Hardware Validation Engineer Apr 2017 – Sep 2021

Amazon Lab126 · Sunnyvale, CA

Led high-speed interface compliance testing (USB, SATA, HDMI) from proof-of-concept through production certification for consumer devices. Owned acceptance criteria, test plans, data collection, and design feedback.

- Developed test automation scripts, data analysis tooling, and repeatable, instrumentation-aware test methodologies.
- Managed test equipment vendor relationships and tracked standards revisions and emerging methodologies.

Senior Hardware Engineer Jul 2014 – Mar 2017

Oracle · Santa Clara, CA

TX/RX high-speed interface characterization: jitter analysis and jitter tolerance, signal quality measurements, PLL tuning, calibration, and S-parameters. Developed test scripts to automate data collection, DUT parameter control, and result formatting.

Senior Hardware Engineer Apr 2008 – Jun 2014

LSI Logic (now Broadcom) · Milpitas, CA

Compliance testing across PCIe, SATA, SAS, Fibre Channel, and IEEE 802.3 standards at speeds up to 32 Gbps. Early adopter of PAM4 benchtop solutions. Contributed data collection and methodology experimentation to the IEEE 802.3bj standards committee. Developed test scripts and equipment drivers to automate measurements, stress-test functionality, and perform statistical analysis.

EDUCATION & RECOGNITION

B.S. Physics, University of California, Irvine 2002 – 2006

Graduated with Honors (3.7 GPA) · UC Regents Scholar · National Physics Honors Society · Campuswide Honors Program

Winner, Document360 Global Writers Awards · recognized for overall body of documentation work 2025

TECHNICAL SKILLS

AI & ML	Large Language Models (LLMs), agentic pipeline architecture, LLM orchestration (Claude, Gemini), prompt engineering, transformer architecture (embeddings, RoPE, Q/K/V attention, backpropagation), Mixture of Experts, RAG, semantic search, vector databases, fine-tuning (LoRA), model evaluation, Model Context Protocol (MCP)
LANGUAGES & TOOLS	Python (NumPy, pandas, matplotlib), Playwright, pytest, YAML, JSON, Markdown, Bash/Zsh, Git/GitHub, Jira
DOCUMENTATION	Technical blog and long-form writing, SME interviews, editing, docs-as-code, GitBook, VuePress, API documentation, information architecture, content strategy, style guides, versioned docs, technical ebooks
DATA & ANALYSIS	Statistical modeling, data visualization (charts, diagrams), parameter sweep optimization, regression analysis, benchmark design, threshold calibration, data pipeline design, privacy modeling (k-anonymity)
INFRASTRUCTURE	Docker, Kubernetes, CI/CD workflows, cloud environments, Linux CLI, headless browser automation, model serving (vLLM, Ollama), AI gateway routing (LiteLLM)